

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Mid-Semester Examination

Course No.: EEE 4483

Course Title: Digital Electronics and Pulse Techniques

Summer Semester, A. Y. 2022-2023

Time: 90 Minutes

Full Marks: 75

There are 3 (three) questions. Answer all 3 (three) questions. The symbols have their usual meanings. Programmable calculators are not allowed. Marks of each question and corresponding COs and POs are written in the brackets. Do not write on the question paper.

1. a) In Fig. 1, an RTL-based NOT gate is working as the driver gate that has a Fan-out equal to N. Calculate the minimum voltage at the input side of each NOT gate guaranteed to be recognized as logic HIGH. Determine the value of N. Describe the effect on Fan-out if a noise of 0.3V is considered. [Here, $V_{BE}(\text{Cut-in}) = 0.65\text{V}$, $V_{BE}(\text{Sat}) = 0.7\text{V}$, $V_{CE}(\text{Sat}) = 0.1\text{V}$, $\beta = 20$, $\sigma = 0.85$] [12] (CO2, PO2)

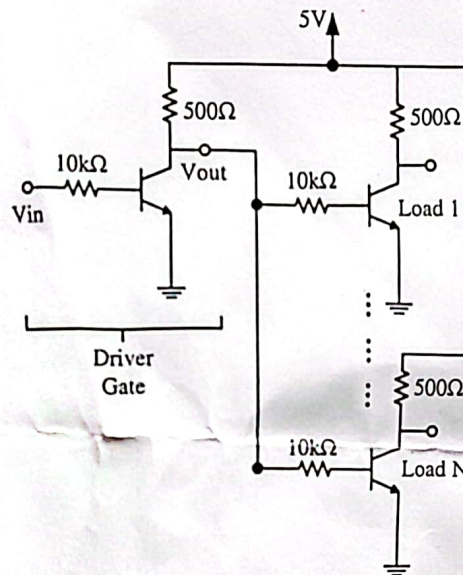


Fig. 1

- b) Identify which type of logic gate is presented in Fig. 2. Considering input V2 to be fixed at 0V, describe the operation of this circuit for the change in input V1 only. [13] (CO2, PO2)
Also sketch the input-output characteristics curve, determine the different input & output voltage levels and noise margins. Consider 0.7 V voltage drop across each diode when it is forward biased. For the transistor, $V_{BE}(\text{Cut-in}) = 0.65\text{V}$, $V_{BE}(\text{Sat}) = 0.7\text{V}$, $V_{CE}(\text{Sat}) = 0.2\text{V}$, $\beta = 50$.

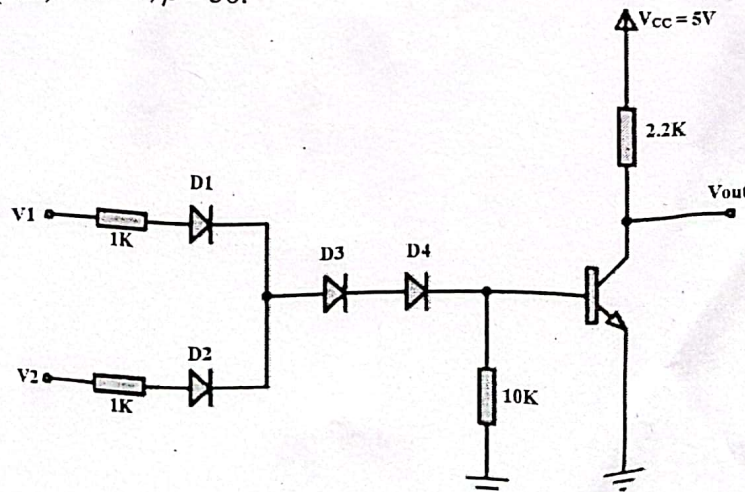


Fig. 2

2. a) Define rise time, fall time and propagation delay related to the operation of logic gates. [6]
(CO1, PO1)
- b) Sketch the physical structure of a multi-emitter transistor and mention the advantage of using this type of transistor. [4]
(CO1, PO1)
- c) For the circuit shown in Fig. 3, find the role of each transistor in the circuit and describe the overall operation of the circuit. For this circuit, sketch the input-output characteristics curve, determine the different input & output voltage levels and noise margins. [Here, $\beta = 50$, $\sigma = 0.85$, $V_{BE}(\text{Cut-in}) = 0.65\text{V}$, $V_{BE}(\text{Sat}) = 0.75\text{V}$, $V_{CE}(\text{Sat}) = 0.2\text{V}$] [15]
(CO2, PO2)

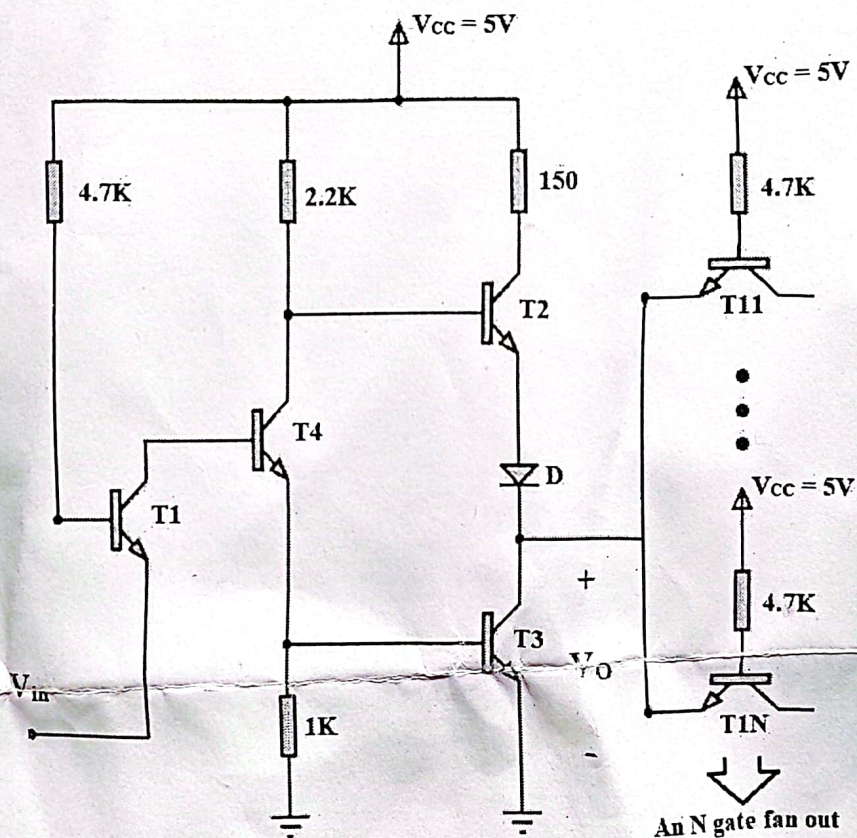


Fig. 3

3. a) Define enhancement type nMOSFET. Describe the operation of enhancement type nMOSFET in all regions of operation and draw its IV characteristic curve clearly showing all regions of operation. [10]
(CO1, PO1)
- b) Sketch the circuit diagram to implement the following Boolean expression using static CMOS logic. [10]
(CO2, PO2)
- $$F = (A + \bar{B}) \cdot C + D \cdot \bar{E}$$
- c) Write some advantages of CMOS inverter design. [5]
(CO1, PO1)